

# Lab: Cognitive Strategies for Inventive Problem Solving

## Objective

Apply four distinct inventive cognition strategies — analogical reasoning, combinatorial creativity, TRIZ contradiction resolution, and mental simulation — to your own invention idea. By comparing the outputs of different cognitive strategies applied to the same problem, you will discover which strategies are most productive for your specific challenge and develop fluency in switching between cognitive modes.

## Materials / Prerequisites

- Your own invention idea or project (any domain, any stage)
- Paper and pen for sketching and diagramming
- A timer
- Completed reading of Module 04: Inventive Cognition

## Part 1: Problem Definition (8 minutes)

Before applying cognitive strategies, clearly define the inventive problem you are working on.

**State your invention problem in one sentence:**

*Write your response here*

**What is the core function your invention must perform?**

*Write your response here*

**What is the primary constraint or trade-off that makes this problem difficult?**

*Write your response here*

**What solution approaches have you already considered?**

*Write your response here*

## **Part 2: Analogical Reasoning (12 minutes)**

Analogical reasoning is the most powerful cognitive tool for invention because it allows you to leverage solutions from well-understood domains to solve problems in less-understood domains. The key to effective analogy is abstraction: you must strip away the surface features of your problem (the specific materials, contexts, and domain jargon) to reveal its structural essence (the relationships, flows, and transformations). Once you have the structural essence, you can search for it in other domains. The best analogies are often surprising — they come from domains that seem completely unrelated on the surface but share deep structural patterns with your problem.

**Step 1 (4 min):** Abstract your problem to its structural essence. Strip away domain-specific details and describe the problem in terms of relationships, flows, transformations, and constraints. For example, "designing a better wheelchair ramp" becomes "facilitating transition between two surfaces at different heights with minimal effort."

Structural abstraction of my problem:

*Write your response here*

**Step 2 (4 min):** Identify three source domains where a structurally similar problem has already been solved. For each, describe the analogy:

| Source Domain | The Analogous Problem | How It Was Solved | Structural Similarity to My Problem |
|---------------|-----------------------|-------------------|-------------------------------------|
|               |                       |                   |                                     |

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**Step 3 (4 min):** Select the most promising analogy and transfer the solution approach to your domain. Be specific about what transfers and what does not:

Selected analogy:

*Write your response here*

What transfers (structural principles):

*Write your response here*

What does NOT transfer (domain-specific details):

*Write your response here*

Resulting invention concept:

*Write your response here*

### **Part 3: Combinatorial Creativity (12 minutes)**

Combinatorial creativity works by assembling components from different generative models into novel configurations. The smartphone combined telephony, computing, photography, navigation, and audio into a single device — none of the components were individually novel, but the combination produced emergent capabilities that none possessed alone. The key to productive combination is selectivity: not all combinations are creative. Random mashing-together produces Rube Goldberg machines, not inventions. Productive combinations require

complementarity (components address different aspects of a coherent need), emergence (the combination creates capabilities that no component has alone), and coherence (the combination can be described by a single unified concept).

**Step 1 (4 min):** List 8-10 existing technologies, methods, materials, or concepts that are individually relevant to your problem domain. Include items from outside your primary domain:

1.

*Write your response here*

1.

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*Write your response here*



**Step 2 (4 min):** Create three novel combinations by pairing items from your list that have not been combined before. For each combination, describe what emergent capability the combination might produce:

| Combination (Item + Item) | Emergent Capability | Is This Complementary? | Is This Coherent? |
|---------------------------|---------------------|------------------------|-------------------|
|                           |                     |                        |                   |

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**Step 3 (4 min):** Select the most promising combination and develop it into a brief invention concept (3-5 sentences):

*Write your response here*

## **Part 4: TRIZ Contradiction Resolution (15 minutes)**

TRIZ approaches invention as a systematic discipline rather than a creative art. At its core, TRIZ identifies the fundamental contradiction in a problem — the conflict between two desirable properties that seem mutually exclusive — and then applies proven resolution

principles to transform the problem so that both properties can be achieved. This transforms the inventor's mindset from "I must choose between A and B" to "I must find a way to have both A and B by changing how the system works." The 40 inventive principles represent decades of analysis across hundreds of thousands of patents, making them an empirically validated toolkit for cognitive transformation.

**Step 1 (5 min):** Identify a core contradiction in your invention problem. A contradiction exists when improving one desirable parameter worsens another.

Parameter I want to improve:

*Write your response here*

Parameter that worsens when I improve the first:

*Write your response here*

State the contradiction in one sentence:

*Write your response here*

**Step 2 (5 min):** Apply three TRIZ inventive principles to resolve the contradiction. For each, describe what it suggests:

| TRIZ Principle                                          | Description | How It Applies to My Contradiction | Resulting Concept |
|---------------------------------------------------------|-------------|------------------------------------|-------------------|
| Segmentation — divide the system into independent parts |             |                                    |                   |

*Write your response here*

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*Write your response here*

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*Write your response here*

| | Dynamism — make a rigid element flexible or variable |

*Write your response here*

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*Write your response here*

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*Write your response here*

| | Inversion — do the opposite of conventional approach |

*Write your response here*

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*Write your response here*

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*Write your response here*

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**Step 3 (5 min):** Which TRIZ principle produced the most promising resolution? Develop it into a brief invention concept:

Selected principle:

*Write your response here*

How it resolves the contradiction:

*Write your response here*

Resulting invention concept:

*Write your response here*

## Part 5: Mental Simulation (13 minutes)

Mental simulation is the cognitive counterpart to physical prototyping. By running your generative model forward in time — imagining the invention in use, predicting how users will interact with it, anticipating failure modes and edge cases — you can identify problems before investing resources in building. Mental simulation is not daydreaming: it is disciplined, step-by-step prediction that deliberately stress-tests the model at each stage. The quality of your mental simulation depends entirely on the accuracy of your generative model. If your model accurately represents the relevant physics, user behavior, and environmental conditions, the simulation will be informative. If your model has gaps or errors, the simulation will be

misleading — which is why alternating between mental and physical prototyping is the most effective strategy.

**Step 1 (3 min):** Choose the most promising invention concept from Parts 2, 3, or 4. Describe it clearly enough to simulate:

Concept to simulate:

*Write your response here*

**Step 2 (5 min):** Run a mental simulation. Imagine the invention exists and is being used. Walk through a complete use scenario, noting what happens at each step:

| Step                                | What Happens | What Could Go Wrong | What Surprised Me |
|-------------------------------------|--------------|---------------------|-------------------|
| User first encounters the invention |              |                     |                   |

*Write your response here*

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*Write your response here*

|



*Write your response here*

**|| User begins using it |**

*Write your response here*

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*Write your response here*

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*Write your response here*

**|| Core function is performed |**

*Write your response here*

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*Write your response here*

|

*Write your response here*

|| User encounters an edge case |

*Write your response here*

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*Write your response here*

|| Long-term use (months later) |

*Write your response here*

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*Write your response here*

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*Write your response here*

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**Step 3 (5 min):** Based on your mental simulation, identify the three biggest risks or uncertainties and propose how to address each:

| Risk / Uncertainty | Severity (High/Med/Low) | How to Test or Resolve |
|--------------------|-------------------------|------------------------|
|                    |                         |                        |

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## Synthesis

You now have three or more invention concepts generated by different cognitive strategies.  
Compare them:

| Strategy                | Concept<br>Generated | Novelty<br>(1-10) | Feasibility<br>(1-10) | My Confidence in<br>It (1-10) |
|-------------------------|----------------------|-------------------|-----------------------|-------------------------------|
| Analogical<br>reasoning |                      |                   |                       |                               |

*Write your response here*

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*Write your response here*

| | **Combinatorial creativity** |

*Write your response here*

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*Write your response here*

**|| TRIZ contradiction resolution |**

*Write your response here*

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Which strategy produced the most useful result? Why do you think that strategy worked best for your particular problem?

*Write your response here*

## **Discussion / Debrief Questions**

1. Did the different cognitive strategies produce genuinely different concepts, or did they converge on similar ideas? What does this tell you about the structure of your problem?
2. Which cognitive strategy felt most natural to you? Which felt most awkward? How does this relate to your inventor archetype from Module 02?
3. The mental simulation step often reveals risks that were invisible during the generative phase. Did your simulation change your assessment of which concept was most promising? How?



4. TRIZ reframes trade-offs as contradictions that can be resolved. Did you find that your problem contained a genuine contradiction, or was the trade-off a result of assumptions that could be questioned?
5. How could you systematically use these four strategies in your regular inventive practice? Would you use them in sequence, in parallel, or selectively based on the problem type?